Kirill Kozlov

Junior Machine Learning Developer (Python, LLM, RAG)

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Professional Summary

Software engineer and Machine Learning master's student at CMC MSU with production experience at X5 Tech. Built Python services using FastAPI, PostgreSQL, and Kubernetes; developing AI product search and catalog-processing pipelines for Korzina.

Technical Skills

Programming Languages: Python, SQL, Go, C++
Artificial Intelligence and Machine Learning: Large Language Models (LLM), LLM agents, RAG (Retrieval-Augmented Generation), BM25, OpenAI API, LangChain, prompt engineering, OCR (Optical Character Recognition), A/B testing, PyTorch, Claude Code
Frameworks: FastAPI, SQLAlchemy (async), Alembic, Pydantic, aiogram, pytest
Libraries: asyncio, httpx, aiohttp, requests, pandas, NumPy, scikit-learn, structlog
Databases and Messaging: PostgreSQL, Redis, Apache Kafka
Infrastructure and DevOps: Docker, Kubernetes (K8s), Helm, GitLab CI/CD, Nginx, Git, Linux
Languages: Russian — Native, English — Upper-Intermediate (B2)

Work Experience

Software Engineer (Junior) | X5 Tech (X5 Group, Russia's largest grocery retailer), Logistics Solutions Department, Moscow | October 2026 — Present

- Developing the system that handles 90% of all foreign supplies X5 Retail Group purchases for its retail chains.

Machine Learning Developer (Intern) | X5 Tech, AI Agents Department, Moscow | April 2026 — October 2026

- Built an LLM agent on LangChain for contract review: over 500 production reviews in May–August 2026, a legal opinion in 8–19 minutes instead of about 97 minutes of manual lawyer work; business impact of 8 FTE for X5 Group.
- Implemented retrieval over a legislation corpus for RAG: 40 federal laws, 366 articles (including the Civil Code), BM25 index, injection of relevant articles into the agent prompt.
- Developed the FastAPI backend for a utility-bill processing service: consuming OCR results from Apache Kafka, storing them and serving them to the frontend; business impact of 10 FTE.
- Built and deployed to dev and prod a foreign invoice analysis service based on OCR and LLM for X5 Tech; in pilot operation since August 2026.
- Deployed the services to a Kubernetes cluster (dev and prod) via Helm and GitLab CI/CD; agent latency is tracked in Grafana.

Projects

Korzina — Marketplace and AI Product Search | TypeScript, Node.js, PostgreSQL, React | September 2026 — Present

- Building a Telegram Mini App and bot with a catalog of 100,000+ products; automated supplier-data ingestion, normalization, and deduplication.
- Implemented PostgreSQL full-text and trigram search; developing a hybrid retrieval prototype combining multimodal Gemini embeddings, pgvector, reciprocal rank fusion (RRF), and Qwen reranking.
- Implemented LLM/VLM attribute extraction and product-compatibility checks in the prototype, with versioned embedding caches, concurrent-request deduplication, and incremental refresh.
- Built offline relevance evaluation with frozen datasets, qrels, Recall@20, and Precision@5; compared text and multimodal retrieval under model-cost and latency constraints.
- Built the Fastify API and background jobs; deployed the service using PostgreSQL, Valkey, S3, Docker, and GitLab CI/CD.

Other projects: AxiomAI: LLM-powered Telegram bot, 80% reduction in manager workload; WB Parser: Wildberries parser and Mini App, 220 users; Placa: analytics for 300 sellers.

Education

Lomonosov Moscow State University, Master's degree | Faculty of Computational Mathematics and Cybernetics, Department of Mathematical Methods of Forecasting, major in Machine Learning, Moscow | September 2026 — Present

Lomonosov Moscow State University, Bachelor's degree | Faculty of Computational Mathematics and Cybernetics, Moscow | August 2022 — August 2026

- Deep Learning School (DLS), Computer Vision and NLP tracks — MIPT School of Deep Learning.
- Summer School on A/B Testing, X5 Tech — Summer 2026 (certificate no. 9074503).